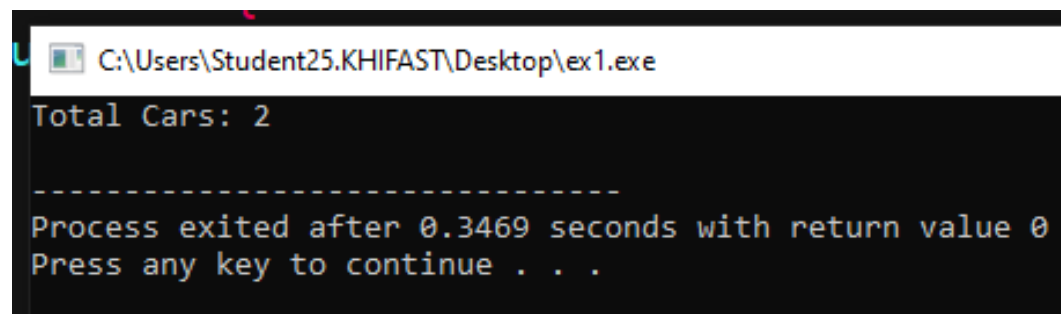


LAB 05

EX 1:

```
1  #include<iostream>
2  #include<string>
3  using namespace std;
4  class Car{
5  public:
6      static int totalCars;
7
8      Car(){
9          totalCars++;
10     }
11 };
12 int Car::totalCars = 0;
13 int main(){
14     Car c1;
15     Car c2;
16     cout<<"Total Cars: "<<c2.totalCars<<endl;
17 }
```

Output:



```
C:\Users\Student25.KHIFAST\Desktop\ex1.exe
Total Cars: 2
-----
Process exited after 0.3469 seconds with return value 0
Press any key to continue . . .
```

EX 2:

```
#include<iostream>
#include<string>
using namespace std;
class BankAccount{
private:
    string accountHolder;
    double balance;
public:
    static int interestRate;

    BankAccount(string name, double bal) {
        accountHolder = name;
        balance = bal;
    }

    static void display(){
        cout << "Current Interest Rate: " << interestRate << "%" << endl;
    }

    void displayAccountInfo() {
        cout << "Account Holder: " << accountHolder << endl;
        cout << "Balance: " << balance << endl;
    }
};
int BankAccount::interestRate= 2;
int main(){
    BankAccount acc1("Ali", 10000);
    BankAccount acc2("Ahmed", 20000);

    BankAccount::display();

    acc1.displayAccountInfo();
    cout << endl;
    acc2.displayAccountInfo();
}
```

Output:

```
Current Interest Rate: 2%
Account Holder: Ali
Balance: 10000

Account Holder: Ahmed
Balance: 20000

-----
Process exited after 0.3601 seconds with return value 0
```

Ex 3:

```
#include<iostream>
#include<string>
using namespace std;
class Book{
    const int ISBN;
public:
    Book(int isbn):ISBN(isbn){}

    void display(){
        cout<<"ISBN: "<<ISBN<<endl;
    }
};

int main(){
    Book b1(10003);
    b1.display();
}
```

Output:

```
C:\Users\Student25.KHIFAST\Desktop\ex3.exe
ISBN: 10003
-----
Process exited after 0.3344 seconds with return value 0
Press any key to continue . . .
```

Ex 4:

```
#include <iostream>
using namespace std;

class Circle {
private:
    double radius;

public:
    Circle(double r) {
        radius = r;
    }
    double area() const {
        return 3.14159 * radius * radius;
    }
};

int main() {

    Circle c1(5);
    cout<<"AREA: "<<c1.area()<<endl;
    return 0;
}
```

Output:

```
AREA: 78.5397
-----
Process exited after 0.3884 seconds with return value 0
Press any key to continue . . .
```

Ex 5:

```
#include <iostream>
using namespace std;

class CPU{
public:
    void boot(){
        cout<<"Computer starting..."<<endl;
    }
};

class Computer{
    CPU a;
public:
    void start(){
        a.boot();
    }
};

int main() {
    Computer c1;
    c1.start();
    return 0;
}
```

Output:

```
Computer starting...
-----
aProcess exited after 0.3472 seconds with return value 0
Press any key to continue . . .
```

Ex 6:

```
#include <iostream>
using namespace std;

class Employee{
    int ID;
    float salary;
public:
    void set(int i,int s){
        ID=i;
        salary=s;
    }
    void display(){
        cout<<"\nID: "<<ID<<"\nSalary: "<<salary<<endl;
    }
};

int main() {
    const int SIZE = 5;
    Employee e[SIZE];

    for (int i = 0; i < SIZE; i++) {
        e[i].set(i + 1,i+100);
    }

    for (int i = 0; i < SIZE; i++) {
        e[i].display();
    }
    return 0;
}
```

```
ID: 1
Salary: 100
```

```
ID: 2
Salary: 101
```

```
ID: 3
Salary: 102
```

```
ID: 4
Salary: 103
```

```
ID: 5
Salary: 104
```

```
-----
```

Output: Process exited after 0.3429 seconds with return value 0

Task 1:

```
#include <iostream>
#include <string>
using namespace std;

class LibraryMember {
private:
    string name;
    static int totalMembers;
public:
    LibraryMember(string name) {
        this->name = name;
        totalMembers++;
    }

    void displayMember() const {
        cout << "Member Name: " << name << endl;
    }

    static void displayTotalMembers() {
        cout << "Total Library Members: " << totalMembers << endl;
    }
};

int LibraryMember::totalMembers = 0;

int main() {
    LibraryMember m1("zain");
    LibraryMember m2("sara");
    LibraryMember m3("ayesha");
    LibraryMember m4("ali");

    m1.displayMember();
    m2.displayMember();
    m3.displayMember();
    m4.displayMember();

    LibraryMember::displayTotalMembers();

    return 0;
}
```

Output:

```
Member Name: zain
Member Name: sara
Member Name: ayesha
Member Name: ali
Total Library Members: 4
```

Task 2:

```
#include <iostream>
#include <string>
using namespace std;

class SchoolAccount {
private:
    string studentName;
    int grade;

public:
    static double tuitionFee;

    SchoolAccount(string name, int g) {
        studentName = name;
        grade = g;
    }

    static void showFee() {
        cout << "Tuition Fee: " << tuitionFee << endl;
    }

    void displayStudent() const {
        cout << "Name: " << studentName << ", Grade: " << grade << endl;
    }
};

double SchoolAccount::tuitionFee = 5000.0;

int main() {

    SchoolAccount s1("Ali", 10);
    SchoolAccount s2("ahmed", 12);

    s1.displayStudent();
    s2.displayStudent();

    SchoolAccount::showFee();
    return 0;
}
```

Output:

```
Name: Ali, Grade: 10
Name: ahmed, Grade: 12
Tuition Fee: 5000

-----
Process exited after 0.1973 seconds with return value 0
Press any key to continue . . .
```


Task 3:

```
#include <iostream>
#include <string>
using namespace std;

class Vehicle {
private:
    const int vehicleID;
    string modelName;

public:
    Vehicle(int id, string model) : vehicleID(id), modelName(model) {}
    void display() {
        cout << "\nVehicle ID: " << vehicleID << endl;
        cout << "Model Name: " << modelName << endl;
    }
};

int main() {
    Vehicle v1(101, "Toyota Corolla");
    Vehicle v2(102, "Honda Civic");

    v1.display();
    v2.display();

    return 0;
}
```

Output:

```
Vehicle ID: 101
Model Name: Toyota Corolla

Vehicle ID: 102
Model Name: Honda Civic

-----
Process exited after 0.1684 seconds with return value
Press any key to continue
```

Task 4:

```
#include <iostream>
using namespace std;

class Square {
private:
    double sideLength;

public:
    Square(double side) : sideLength(side) {}

    double calculateArea() const {
        return sideLength * sideLength;
    }

    void display() {
        cout << "Side Length: " << sideLength
              << ", Area: " << calculateArea() << endl;
    }
};

int main() {
    Square s1(5.0);
    Square s2(7.5);

    s1.display();
    s2.display();

    return 0;
}
```

Output:

```
Side Length: 5, Area: 25
Side Length: 7.5, Area: 56.25

-----
Process exited after 0.1711 seconds with return value 0
Press any key to continue . . .
```

Task 05:

```
#include<iostream>
#include<string>
using namespace std;
class SIM{
    string carrier;

    public:
    SIM(){}
    SIM(string c){
        carrier=c;
    }
    void showCarrier(){
        cout<<"Carrier: "<<carrier<<endl;
    };
};
class Mobile{
    string model;
    public:
    SIM s;
    Mobile(string m,string c):s(c){
        model=m;
    }
    void showModel(){
        cout<<"Model: "<<model<<endl;
        s.showCarrier();
    }
};
int main(){
    Mobile m1("A2+","telenor");
    Mobile m2("S24","zong");
    m1.showModel();
    m2.showModel();
}
```

Output:

```
rumaisa-abbasi/Desktop/classes/"task5
Model: A2+
Carrier: telenor
Model: S24
Carrier: zong
rumaisa-abbasi@abdullah-muaz-ThinkPad-L1
```

Task 6:

```
#include<iostream>
#include<string>
using namespace std;
class Product{
    int ID;
    string name;
    int price;
public:
    void input() {
        cout << "Enter ID: ";
        cin >> ID;
        cout << "Enter name: ";
        cin>>name;
        cout << "Enter price: ";
        cin >> price;
    }
    void display(){
        cout<<"Product ID: "<<ID
        <<"\nName: "<<name<<"\nPrice:
        "<<price<<endl;
        cout<<"-----
        -----"<<endl;
    }
};
int main(){
    Product arr[4];
    for(int i=0;i<4;i++){
        arr[i].input();
    }cout << "\nAll Product Details:" << endl;
    for(int i=0;i<4;i++){
        arr[i].display();
    }
}
```

Output:

```
Enter ID: 101
Enter name: Applejuice
Enter price: 120
Enter ID: 102
Enter name: mangojuice
Enter price: 150
Enter ID: 103
Enter name: orangejuice
Enter price: 200
Enter ID: 104
Enter name: chocolatemilk
Enter price: 300
```

All Product Details:

Product ID: 101
Name: Applejuice
Price: 120

Product ID: 102
Name: mangojuice
Price: 150

Product ID: 103
Name: orangejuice
Price: 200

Product ID: 104
Name: chocolatemilk
Price: 300

Task 7:

```
#include <iostream>
#include <string>
using namespace std;

class Patient {
public:
    const int patientID;
    string name;
    Patient(int id, string n) : patientID(id), name(n) { }
};

class Hospital {
    Patient* arr[10];
    static int totalPatients;
public:
    Hospital() { }
    void addPatient(int id, string n) {
        arr[totalPatients] = new Patient(id, n);
        totalPatients++;
    }
    void display() {
        for (int i = 0; i < totalPatients; i++) {
            cout << "Patient ID: " << arr[i]->patientID
                 << "\nName: " << arr[i]->name
                 << "\n-----" << endl;
        }
    }
};

int Hospital::totalPatients = 0;
int main() {
    Hospital h;
    h.addPatient(101, "Ali");
    h.addPatient(102, "Sara");
    h.addPatient(103, "Ahmed");
    h.display();
}
```

Output:

```
Patient ID: 101
Name: Ali
-----
Patient ID: 102
Name: Sara
-----
Patient ID: 103
Name: Ahmed
-----
```